

SZÉCHENYI ISTVÁN EGYETEM

MBT and MBTL Determination from Cylinder Pressure Data

Operating Point Recommendations for Endurance and Drag Racing Applications

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1. Brief

Analyzed cylinder pressure data from a prior test bench session using the BDN combustion analyser, determining Minimum Spark Advance for Best Torque (MBT) and knock-limited spark advance (MBTL) across a range of air-fuel ratios, then recommended optimal operating points for endurance racing and drag racing applications.

2. MBT and MBTL: Definition and Engineering Significance

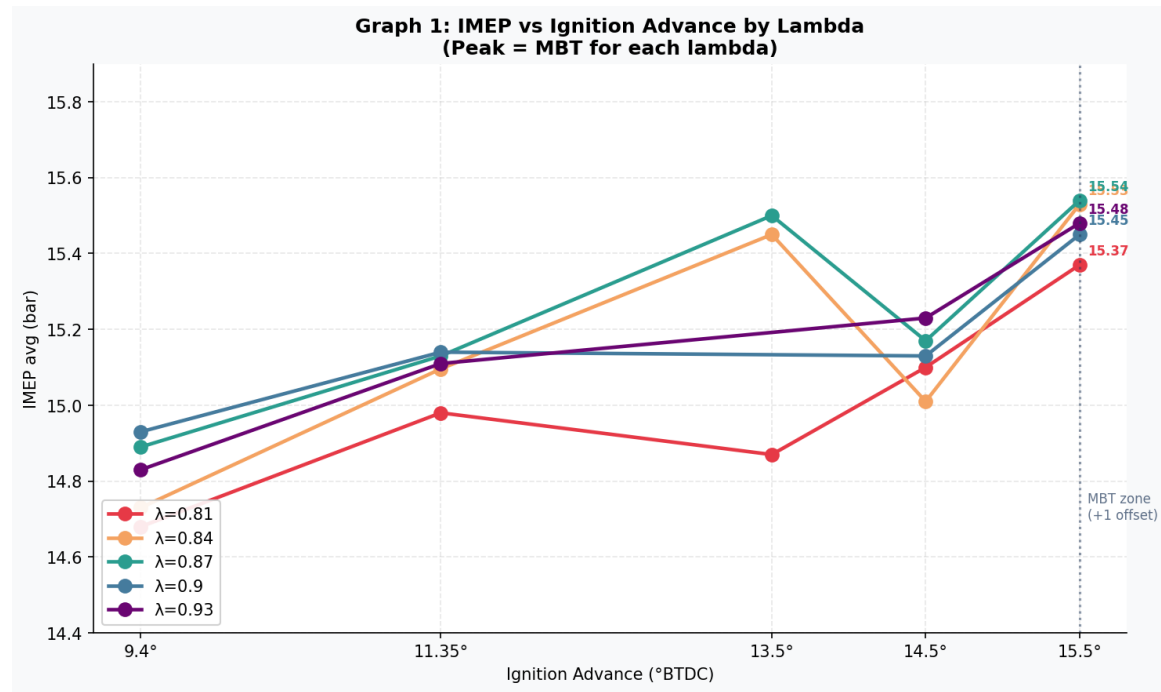
MBT (Minimum Spark Advance for Best Torque) is the ignition timing at which the engine produces its maximum torque output for a given operating condition. It represents the optimal balance between two competing effects: advancing the spark too early causes excessive negative work before TDC as pressure builds against the rising piston, while retarding it too late means peak pressure occurs too far into the expansion stroke, reducing work extraction efficiency. The combustion analyser confirms MBT when MFB50 falls between 7–15° ATDC and gross IMEP reaches its peak value.

MBTL (MBT at the Knock Limit) is the practical ceiling of spark advance in real-world engine operation. When the engine begins to detonate before true MBT timing is reached, the knock-limited advance becomes the safe operating boundary. This is identified from the cylinder pressure trace as the timing at which high-frequency pressure oscillations first exceed the acceptable knock amplitude threshold — defined as 1 bar per 1000 RPM. Beyond this point, the risk of engine damage becomes significant.

Together, MBT and MBTL define the ignition timing operating window for any given lambda and load condition. Understanding both is essential for maximising performance while protecting engine integrity, and their relationship shifts with air-fuel ratio — richer mixtures are generally more knock-resistant, allowing timing closer to true MBT, while leaner mixtures narrow this window considerably.

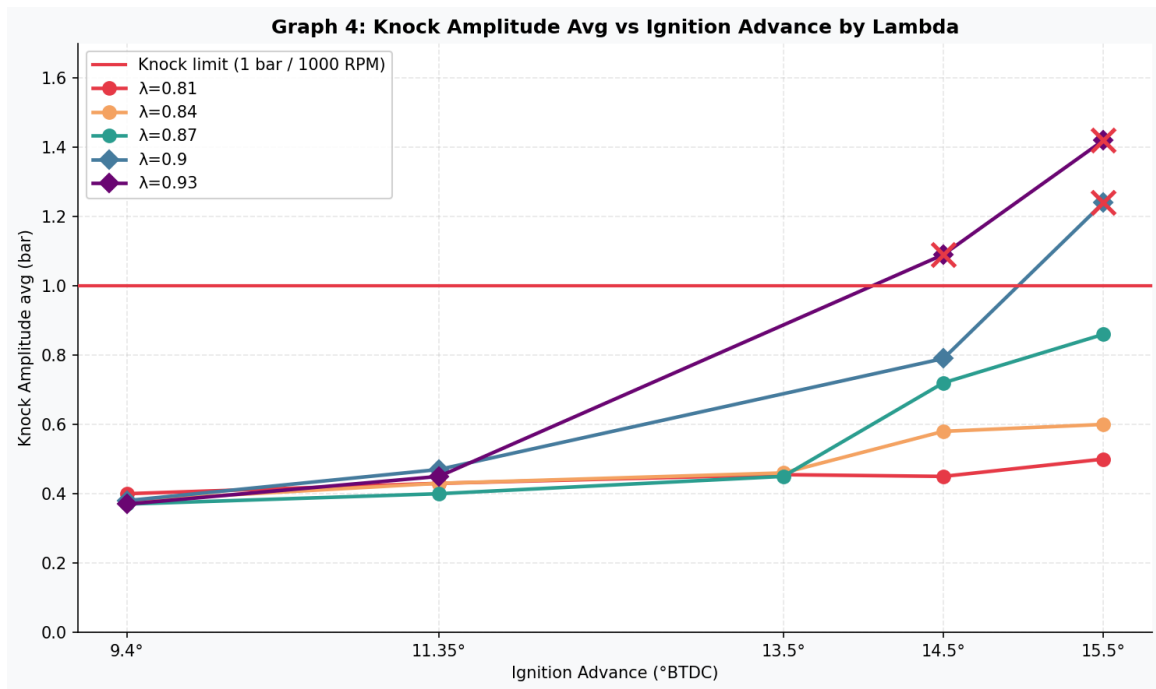
3. Test bed results

In order to find MBT and MBTL we must analyse the results from the dyno runs, using BDN combustion analyser and Ignitron engine management system.



1. Figure: IMEP vs Ignition Advance by Lambda

Indicated Mean Effective Pressure is the most direct measurable indicator of engine torque output from cylinder pressure data, making it the primary parameter for identifying MBT. All five lambda values reach their highest IMEP at the most advanced timing of 15.5°BTDC (+1 in the basic matrix that was given for this task), with $\lambda=0.87$ and $\lambda=0.84$ producing the peak values of 15.54 and 15.53 bar respectively, identifying these as the strongest MBT candidates. As ignition timing is retarded toward 9.4°BTDC, IMEP drops consistently across all lambda values, confirming that combustion phasing is moving away from the optimal expansion window. A notable finding is that IMEP remains relatively stable across different lambda values at the same advance timing, indicating that mixture strength has less influence on torque output than ignition timing within this operating range. **For MBT the best log file was the number 12 → +1°Ignition advance and 0,87 λ .**



2. Figure Knock Amplitude Avg vs Ignition Advance by Lambda

This graph is the primary tool for identifying MBTL, showing how average knock amplitude evolves with ignition advance across all five lambda values relative to the 1 bar per 1000 RPM safety limit shown as the red reference line. Lambda values of 0.90 and 0.93 clearly exceed this limit at 15.5°BTDC advance, marked with cross symbols, confirming these operating points are knock limited before true MBT is reached. The three richer mixtures of $\lambda=0.81$, 0.84, and 0.87 remain safely below the knock limit across all tested advance values, demonstrating sufficient knock resistance to operate at MBT timing. The divergence between lean and rich mixtures is most pronounced at the highest advance setting, consistent with the principle that leaner mixtures burn slower and hotter, increasing auto-ignition probability. **Log 13 is the clearest MBTL example at +1° Ignition advance and 0,9 λ , reaching 1.24 bar knock avg.**

4. Drag Racing setup

Bottlenecks and Key Factors in Drag Racing

Drag racing places fundamentally different demands on an engine compared to endurance racing. The entire performance window is compressed into a matter of seconds, meaning the priority shifts completely away from reliability and fuel economy toward extracting maximum power output at every moment of the run. The two primary bottlenecks in drag racing engine setup are traction management at launch and peak torque delivery through the acceleration phase, with the engine calibration playing a critical role in both.

For the engine setup specifically, the most important factors are maximum IMEP and torque output, optimal combustion phasing as close to MBT as possible, and sufficient mixture richness to protect the engine thermally during the high load event. Knock tolerance is significantly less critical than in endurance racing because the short duration of a drag run means even elevated knock events do not have sufficient time to cause cumulative mechanical damage, provided they remain within reasonable bounds.

Why Log 11 is the Best Choice

Log 11 at $\lambda=0.84$ with 15.5° ignition advance produces the highest IMEP of 15.53 bar, peak torque of 197 Nm and 54 kW @ 2600 RPM (this does not represent peak engine power, as the measurement sweep was conducted at a fixed steady-state RPM for ignition timing analysis purposes). Among all tested operating points, making it the strongest performer in pure output terms. The ignition advance of 15.5° BTDC is the most aggressive setting in the dataset, placing combustion phasing at MFB50 of 14.98° which sits right at the boundary of the optimal $7\text{--}15^\circ$ ATDC window, confirming that the combustion event is well timed for maximum work extraction. The lambda value of 0.84 provides a slightly leaner mixture than the richest tested point while still delivering superior torque output, demonstrating that the additional fuel energy at $\lambda=0.81$ does not translate into better performance at this operating condition. Peak cylinder pressure of 64.23 bar and average knock amplitude of 0.60 bar both confirm a controlled and efficient

combustion event, and the EGT of 835°C remains well within safe limits even for a high intensity short duration run. This combination of maximum torque, optimal combustion phasing, and controlled thermal loading makes **Log 11 the clear recommendation for drag racing application on this engine.**

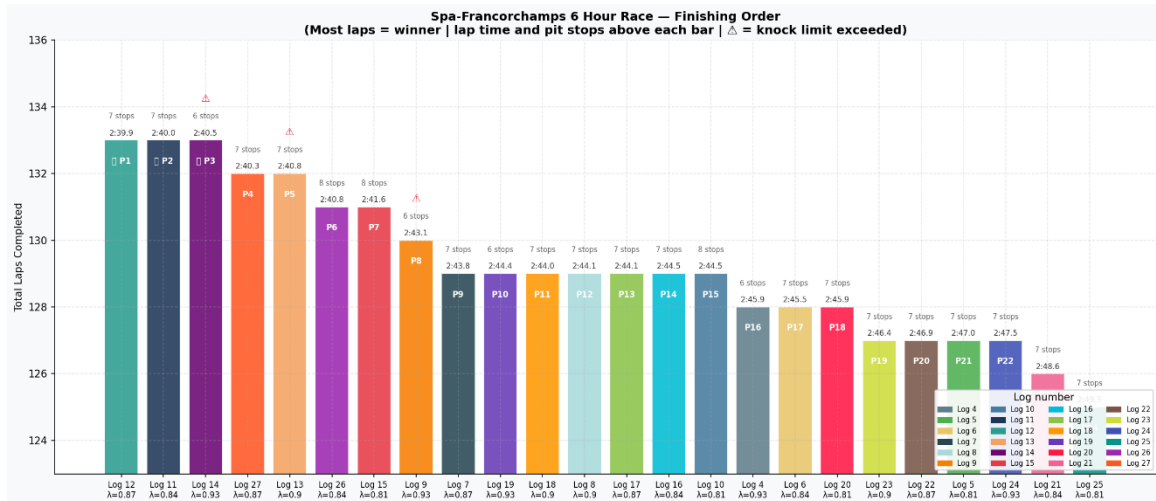
5. Endurance racing setup

The fundamental difficulty of endurance racing engine calibration is finding the optimal balance between performance and fuel economy. Unlike drag racing where maximum power is the sole objective, every additional litre of fuel consumed translates directly into more pit stops, and each pit stop costs track time that cannot be recovered. A setup that produces marginally more torque but forces one extra fuel stop will almost certainly finish behind a more economical setup, making fuel consumption the dominant strategic variable. At the same time the engine must remain reliable across hundreds of combustion cycles under sustained high load, meaning knock amplitude, exhaust gas temperature, and combustion stability all impose hard limits on how aggressively the engine can be tuned.

Simulation parameters

- Track: Spa-Francorchamps, Belgium
- Track length: 7.004 km
- Elevation change: 102 m
- Race duration: 6 hours
- Reference lap time: 2:40 (160 seconds)
- Fuel tank capacity: 55 litres
- Pit stop time loss: 68 seconds per stop (refuel to full each stop)
- Fuel consumption correction factor: 1.35× applied to steady-state measured values to account for dynamic circuit conditions
- Lap time scaling: proportional to IMEP relative to the reference operating point

With these parameters, the following are the results:



3. Figure Simulation Finishing Order

From figure 3 it can be clearly seen that logs 11 and 12 are really close at the end, and in this case the best strategy is to have a bit stronger performance, rather than better fuel economy. Log 14 pitted only six times — one fewer than the top two — but that operating point exceeds the knock limit, risking a DNF. The worst operating point is at $\lambda=0,81$ and ignition advance -5, at this point the car uses a lot of fuel but does not have proper engine performance output.

Appendix: Log numbers mapped to their corresponding ignition advance and lambda settings from the test bench session.

SIMA	0,81	0,84	0,87	0,9	0,93	0,96	0,99
+1	15	11	12	13	14		
0	10	6	7	8	9		
-1	5	1 ²⁶	2 ²⁴	3 ²⁸	4		
-3	20	16	17	18	19		
-5	25	21	22	23	24		